

**Divergence, Sink Contraction, and the Half-Phase Geometry:
Charge, the Antigravity Field, the Photon, and Spin 1/2 from
Flowing Space**

Yuan-Jie Liu*

(Dated: October 7, 2026)

Abstract

We extend the flowing-space framework in two geometric directions and join them into one chain.

(i) *The half-phase of the critical sheet*: the inversion fixed point of the envelope reflection $w \mapsto e/w$ sits at $|w| = \sqrt{e}$, so $\ln(\sqrt{e}) = 1/2$; the square root is the half-step operator of the multiplicative group, and in phase space the same half-step is the half-angle phase $\exp(i\theta/2)$, whose closure factor is -1 — the explicit fermionic witness of spin $1/2$. (ii) *Charge as flow divergence*: positive charge is a source (positive divergence), negative charge a sink (negative divergence), mutually exclusive, flipped under $\rho \rightarrow -\rho$, neutral at zero. A time-varying source forces the space field to vary (CR1); a varying field activates the nuclear channel $m dC$ of the four-force decomposition (CR2); and a sink — a converging flow — is geometrically a flattening: the inward contraction $v_i \mapsto (1-\lambda)v_i + \lambda\bar{v}$ reduces the fluctuation energy Q_A by the square factor $(1-\lambda)^2$ (CR9, exact discrete theorem). Joining the framework’s own μ -bridge (one flattening gives $\mu = 1$), the chain closes: accelerated charge $\Rightarrow$ antigravity field $\Rightarrow$ the photon (mass zero and neutral). As a device-level application, the self-flattening of circulating electrons shortens the approach to the working window $\mu = 0.999$ from 132 to 85 steps and reaches it with halved external RMF drive. All statements are formalized in Lean 4 with zero `sorry`; the honest status is stated: λ (the sink contraction rate) and δ, ε (the sink energy scale) are inputs, and the continuum 3D divergence-to-functional correspondence is not yet formalized.

“Flowing water does not fight; it yields, and in yielding it reaches everywhere.”

— *after Daodejing*

I. INTRODUCTION

The flowing-space framework [1, 2] adopts a single primitive — *flow* — and derives mass, information, causality, force, the photon, and the quantum relations from a small set of postulates: vector light speed, three anchored directions, and the counting of space-displacement strands. The present work adds two geometric structures that were absent there: the *half-phase geometry* of the critical sheet, and the *divergence structure* of charge as source and sink.

* yuanjieliu65@gmail.com

The two additions are independent in content but share a common theorem-level fact: a quadratic quantity is reduced by a square factor. For spin, the phase closure factor of the half-angle map is $-1 = (-1)^2$ -free: the half step of the variable θ is the square root in logarithmic coordinates, and the fermionic witness is $\exp(i\theta/2)$. For charge, the fluctuation energy of a converging flow is reduced by $(1 - \lambda)^2$ per contraction step; the sink is a self-flattening structure, and a sink at full strength ($\lambda \rightarrow 1$) brings the fluctuation to zero, mass to zero, and charge to neutrality — the photon.

II. POSTULATES OF THE FLOWING-SPACE FRAMEWORK

We recall the postulates needed here; full derivations are in Ref. [1].

Postulate II.1 (Vector light speed). *Space itself flows at the vector speed C with $|C| = c$; the direction of C is changeable.*

Postulate II.2 (Three anchored directions). *Space has three independent orthogonal directions; a physical object is an anchored pattern of the flow, with anchor strength $s \geq 0$. The anchored mass is*

$$m_{\text{eff}}^2 = s^2(1 - \mu)^2, \quad (1)$$

where $\mu \in [0, 1]$ is the antigravity strength (flattening of the flow gradient); $\mu = 1$ gives $m_{\text{eff}} = 0$ (MassCancellation AMC1).

Postulate II.3 (Four forces as Leibniz channels). *The flow momentum is $P = m(C - v)$; its time derivative decomposes into four channels (GQF2):*

$$\frac{d}{dt}[m(C - v)] = \underbrace{\dot{m}C}_{\text{electric}} + \underbrace{m\dot{C}}_{\text{nuclear}} - \underbrace{\dot{m}v}_{\text{magnetic}} - \underbrace{m\dot{v}}_{\text{gravity}}. \quad (2)$$

III. CHARGE AS FLOW DIVERGENCE

Charge is the divergence of the space-displacement flow (GQF4; VibrationChargeFlow CF1–CF5).

Theorem III.1 (Source and sink). *For a local divergence scalar ρ :*

$$\text{PositiveSource}(\rho) : \Longleftrightarrow 0 < \rho, \quad (3)$$

$$\text{NegativeSink}(\rho) : \Longleftrightarrow \rho < 0. \quad (4)$$

Positive charge is a source (diverging displacement), negative charge a sink (converging displacement); both obey the right-handed helical motion of the flow, the sign of the divergence distinguishing them.

Theorem III.2 (Source–sink exclusivity; CF1–CF4). *(i) A source and a sink cannot hold simultaneously. (ii) Charge conjugation is the sign flip: $0 < \rho \iff \rho < 0$ after $\rho \mapsto -\rho$ (source and sink are exchanged). (iii) Zero divergence is neutral: neither source nor sink. (iv) The magnitude is preserved under the flip: $|\rho| = |-\rho|$.*

The continuity side — the coupled extraction of the charge e , the divergence dynamics in 3D, and charge normalization — remains open (honestly registered in the repository).

IV. ACCELERATED CHARGE IMPLIES THE FIELD VARIES; THE NUCLEAR CHANNEL ACTIVATES

Theorem IV.1 (CR1; time-varying source forces a varying field). *In the sourced wave equation $\partial_t^2 C = c^2 \partial_x^2 C + J$ (MS5), a static field implies a static source: if $\partial_t C \equiv 0$, then $J(t+1) = J(t)$ for all t . Hence a time-varying source (an accelerated charge) forbids a static space field.*

Theorem IV.2 (CR2; the nuclear channel activates). *With $m \neq 0$ and $\dot{C} \neq 0$, the nuclear channel of Eq. (2) is non-zero: $m \dot{C} \neq 0$.*

The missing link between the nuclear channel and the antigravity parameter μ is not invented here: the framework already contains its own μ -bridge.

Theorem IV.3 (CR3; one flattening gives $\mu = 1$). *Let $Q_A(v) = \sum_{i \in A} (v_i - \bar{v}_A)^2$ be the fluctuation energy, and $\eta = 1 - Q_A(v)/Q_A(v_0)$ the flattening progress (MuFieldCoupling TD11). The μ update is $\mu' = \mu + \eta(1 - \mu)$ (PlasmaDynamics muStep). Flattening once gives $\mu' = \mu + 1 \cdot (1 - \mu) = 1$.*

Theorem IV.4 (CR4; $\mu = 1$ is the photon endpoint). *At $\mu = 1$, the anchored mass vanishes (Eq. 1 with $1 - \mu = 0$) and the divergence is neutral (CF3). The object is massless and chargeless — the photon.*

Theorem IV.5 (CR5; closure of the chain). *One flattening applied to any region A (any anchored substance s) yields a photon: mass zero and neutrality.*

V. CONVERGENCE IS FLATTENING: SINK CONTRACTION

Why is the *negative* charge the natural antigravity carrier? The geometric answer: a sink is a converging flow, and convergence is flattening.

Theorem V.1 (Sink contraction; CR9). *For a region A , a field $v : A \rightarrow \mathbb{R}$, and $\lambda \in [0, 1]$,*

$$\text{sinkContract}(A, v, \lambda)(i) = \begin{cases} (1 - \lambda) v_i + \lambda \bar{v}_A & i \in A, \\ v_i & i \notin A. \end{cases}$$

Each point of A is pulled toward the region mean by a fraction λ — the discrete image of “the negative charge absorbs the surrounding flow toward itself.”

Theorem V.2 (CR9; square decay of fluctuation). *The contraction preserves the region mean and scales the fluctuation energy by $(1 - \lambda)^2$:*

$$Q_A(\text{sinkContract}(A, v, \lambda)) = (1 - \lambda)^2 Q_A(v). \quad (5)$$

For $0 < \lambda < 1$ the decay is monotone and exact; $\lambda = 1$ is a single flattening (fluctuation to zero). The sink is a self-flattening operator family.

Theorem V.3 (CR10; strict attraction pointwise). *For $0 < \lambda < 1$ and $v_i \neq \bar{v}_A$,*

$$|\text{sinkContract}(A, v, \lambda)(i) - \bar{v}_A| < |v_i - \bar{v}_A|,$$

each point is strictly attracted to the mean — “the sink pulls the surrounding flow to itself,” pointwise.

The energy side of the sink is already present in the framework: information sinking into the absorbing branch iR releases $\varphi(R) - \varphi(iR) = \delta + \varepsilon > 0$ (HiddenQFT HQ9a), so a converging flow carries its own flattening power as a candidate (CR8). This is a semantic correspondence (interface), not a theorem of the divergence dynamics.

VI. THE HALF-PHASE GEOMETRY: SPIN 1/2 FROM THE CRITICAL SHEET

The second geometric addition concerns the value 1/2 itself.

A. The inversion fixed point

The Riemann–Hilbert envelope of the framework [3] carries the reflection $s \mapsto 1 - s$; in the logarithmic coordinate $w = e^s$ this is the inversion $w \mapsto e/w$. Its fixed set is the circle $|w| = \sqrt{e}$ (envelope inversion fixed circle). Taking logarithms:

$$\ln(\sqrt{e}) = \frac{1}{2} \ln e = \frac{1}{2}. \quad (6)$$

Theorem VI.1 (CS2; inversion fixed iff log half). *For $r > 0$,*

$$r = e/r \iff \ln r = \frac{1}{2}.$$

The fixed point of the reflection must sit at $1/2$ in the logarithmic coordinate — because $\ln e = 1$ and the fixed point of $s \mapsto 1 - s$ is $1/2$.

B. The half-angle phase is fermionic

The square root is the half-step operator of the multiplicative group; in the phase space of a vibration, the same half-step is the half-angle phase.

Theorem VI.2 (CS4; half-angle phase has closure factor -1). $\theta \mapsto \exp(i\theta/2)$ *satisfies*

$$f(\theta + 2\pi) = -1 \cdot f(\theta),$$

i.e. it is a fermionic closure witness (VibrationStatistics ClosureFactor with $\sigma = -1$): 2π changes sign, 4π restores. The half-angle phase is the explicit witness of spin $1/2$ in the phase space of a vibration.

C. Energy and mass

With $E \propto m^2$ (mass–energy quadrature), logarithms are linear: $\ln E = 2 \ln m$. At the critical sheet, $\ln(\sqrt{e}) = 1/2$ means $m = \sqrt{e}$, $E = e$; the factor 2 in $\ln E = 2 \ln m$ is the inverse of the spin half-step — energy takes two half-steps to complete one cycle of m .

Theorem VI.3 (CS5; logarithmic 2:1). $\ln e = 2 \ln(\sqrt{e})$.

VII. DEVICE APPLICATION: TWO-FLOW-RING OPTIMIZATION

The sink-flattening theorem (CR9) enters the μ dynamics of the two-flow-ring device [4] as a self-flattening gain:

$$\eta_{\text{sink}}(\lambda) = 2\lambda - \lambda^2, \quad (7)$$

the fraction of fluctuation removed by one contraction, added to the external RMF gain and decaying with the residual $\eta_{\text{sink}} \cdot (1 - \mu)$ (the flattening loses its object as the region flattens; TD18). The comparison (evaluated numerically; honest candidate, λ an input):

TABLE I. μ trajectory to the working window $\mu_{\text{work}} = 0.999$ (FC11 ceiling 0.999780568).

	Pure external RMF	External + self-flattening ($\lambda = 0.3$)
Steps to $\mu = 0.999$ (CR1 criterion)	132	85
Steps with halved external drive	267	150

The self-flattening shortens the approach to the working window by $\sim 35\%$ and reaches it with halved external RMF drive — the sink’s own converging geometry contributes to the μ production (supply-side, CR8 candidate). The negative result is registered honestly: a *constant* η_{sink} violates the FC11 ceiling (pushes μ to 1); the decaying form is required. With a constant external drive, μ still tends to 1 (a known TM-family behavior); stopping inside the window requires feedback shutoff (TD19), a separate matter.

VIII. NUMERICAL VERIFICATION

All theorems of the chain are verified in Lean 4 with zero `sorry` (4348 jobs, full build green) and numerically at machine precision where a numeric witness exists:

- 3D 21^3 lattice, negative-divergence field $v = -r$: every interior point has $\text{div } v < 0$; initial fluctuation $Q = 10187$; 200 sink-contraction steps drive $Q \rightarrow 3.5 \times 10^{-61}$ with $|\text{div}| \rightarrow 0$, monotonically (verify_sink_contract_3d).
- Sink contraction scaling $Q \rightarrow (1 - \lambda)^2 Q$ reproduced exactly (CR9 witness); strict pointwise attraction (CR10) monotone in λ .

- Two-flow-ring comparison (Tab. I): $132 \rightarrow 85$ steps at equal external drive; $267 \rightarrow 150$ with halved external drive; $\eta_{\text{sink}}(\lambda)$ strictly increasing in λ (verify_sink_mu_optimization).

IX. HONEST ASSESSMENT

a. Claimed. (i) Charge is the divergence of the flow: source = positive divergence, sink = negative divergence, with the four algebraic facts (exclusivity, flip, neutrality, magnitude) formalized. (ii) The chain “accelerated charge $\Rightarrow$ varying field $\Rightarrow$ nuclear channel $\Rightarrow$ one flattening $\Rightarrow \mu = 1 \Rightarrow$ photon” is a closed conditional chain of theorems: *if* a flattening is applied, any anchored substance becomes a photon. (iii) A sink is geometrically a self-flattening operator: the exact discrete theorem $Q_A \rightarrow (1 - \lambda)^2 Q_A$ under sink contraction. (iv) The half-step geometry locates spin $1/2$: $\ln(\sqrt{e}) = 1/2$ is the fixed point of the envelope reflection in the logarithmic coordinate, and the half-angle phase is the explicit fermionic closure witness. (v) All statements are formalized in Lean 4 with zero `sorry`; the device comparison is numerically verified.

b. Not claimed. (i) The bridge “nuclear channel $\Rightarrow$ flattening action” is *not* derived: who applies the flattening (the physical origin of the flattening power) is the second-input gap, unchanged. (ii) λ (the sink contraction rate) and δ, ε (the sink energy scale) are inputs, not derived values; $\eta_{\text{sink}} = 2\lambda - \lambda^2$ is a candidate implementation, not the unique mapping. (iii) The continuum 3D correspondence “negative divergence $\Rightarrow$ fluctuation functional decreases” is not formalized (the discrete lattice theorems are the algebraic seed). (iv) Spin $1/2$ is *located* geometrically, not *derived* as a value: the other half of the spin–statistics theorem (phase $\Rightarrow$ operator anticommutation) requires field-operator language absent from the framework — the result is half a theorem. (v) No new falsifiable prediction beyond the standard framework; all numerical values reproduce known physics.

X. CONCLUSION

Two geometric structures extend the flowing-space framework. The divergence structure of charge makes the sink a self-flattening operator, closing — conditionally — the chain from accelerated charge to the photon and giving the two-flow-ring device a self-flattening gain that shortens the approach to the working window by a third. The half-phase geometry

locates spin $1/2$ at the envelope-reflection fixed point $\ln(\sqrt{e}) = 1/2$, with the half-angle phase as its explicit fermionic witness. Both additions are theorem-exact where they are exact, and honestly open where the inputs remain: λ , δ , ε , and the continuum 3D correspondence.

-
- [1] Y.-J. Liu *et al.*, “The geometric description of physics at the cosmic scale,” companion manuscript (paper/), repository `logos-42/Hibs-Physics`.
 - [2] `logos-42/Hibs-Physics` repository: Lean 4 formalization and numerical verification, <https://github.com/logos-42/Hibs-Physics>.
 - [3] `logos-42/RiemannHIBS`: hidden-number envelope with the reflection $s \mapsto 1-s$ and the inversion fixed circle $|w| = \sqrt{e}$.
 - [4] Y.-J. Liu, “Two-flow-ring design: gap-tolerant feasible region,” *Design manual* of the repository (`docs/wiki/design-two-flow-ring.md`), and the companion device project `logos-42/HushFusion`.